Computer Lab 4
Bioinformatics

Linköpings Universitet, IDA, Statistik

2026 XII 8

Kurskod och namn:	732A51 Bioinformatics
Datum:	2026 XII 07—2026 XII 12 (lab session 08 XII 2026 SU17,18)
Delmomentsansvarig:	Krzysztof Bartoszek, Hirbod Gholamniaetakhsami
Instruktioner:	This computer laboratory is part of the examination for the Bioinformatics course Create a group report, on the solutions to the lab as a **.PDF** file. Be concise and do not include unnecessary printouts and figures produced by the software and not required in the assignments. **All R code should be included as an appendix into your report.** In the report reference **ALL** consulted sources and disclose **ALL** collaborations. The report should be handed in via LISAM (or alternatively in case of problems e–mailed to hirbod.gholamniaetakhsami@liu.se), by **23:59 12 December 2026** at latest. Notice there is a deadline for corrections **23:59 1 February 2027** and a final deadline of **23:59 1 March 2027** after which no submissions nor corrections will be considered and you will have to redo the missing labs at the next course opportunity. The report has to be written in English.

This assignment is based on the material provided from the R Bio: Untangling Genomes course, by Martin Morgan, Houtan Noushmehr, Tathiane Maistro Malta and others that took place in Montevideo (2015.10.05 – 2015.10.10)
https://www.bioconductor.org/help/course-materials/2015/Uruguay2015/ . You should go through all the material of this tutorial. The following page,
https://gist.github.com/hckiang/ba6bdabdcd0b401c29077d752d6cc11c , might be useful in case of problems with installing some of the required software.

On the webpage https://www.bioconductor.org/help/course-materials/2015/Uruguay2015/day3-gene.expression.html you can find a step by step analysis of gene expression data from HUVEC [1] and Ocular Vascular Endothelial[2] Cells. The data itself (as indicated in the analysis) can be found at ftp://ftp.ncbi.nlm.nih.gov/geo/series/GSE20nnn/GSE20986/suppl/ . The description of the data can be found at https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE20986 .

It might be helpful if you also go through `https://www.bioconductor.org/help/course-materials/2015/Uruguay2015/day5-data_analysis.html` and
https://www.bioconductor.org/help/course-materials/2015/Uruguay2015/V6-RNASeq.html . The latter for graphic producing code.

Question 1

Run all the R code and reproduce the graphics. Go carefully through the R code and explain in your words what each step does.
HINT Recall what a design/model matrix is from linear regression.

Question 2

In the presented analysis, there are no plots of raw paired data. In the section where the contrasts are defined find the three contrasts. Present the variables versus each other original, log–scaled and MA–plot for each considered pair both before and after normalization. A cluster analysis is performed on the page but not reported. Present plots and also draw heatmaps.

Question 3

The volcano plot is only for `huvec` versus `choroid`. Provide volcano plots for the other pairs. Indicate significantly differentially expressed genes. Explain how they are found.

Question 4

Try to find more information on the genes that are reported to be significantly differentially expressed. The place to start off is https://www.ncbi.nlm.nih.gov/gene/, remember that the data is from the species human. Try to look also for other databases where (some) information on the genes may be found. Try to follow on some of the provided links. Report in your own words on what you find.

[1] `https://en.wikipedia.org/wiki/Human_umbilical_vein_endothelial_cell`
[2] `https://en.wikipedia.org/wiki/Endothelium`

Report all the Gene Ontology (GO) terms associated with each gene. Are any of the GO terms common between genes? If so do the common GO terms seem to be related to anything particular? Try to present GO analysis in an informative manner, if possible visualize.